

$$m(\dot{v}_x - v_y r) = F_x - \frac{1}{2}\rho C_d A v_x^2 - F_{rr} - F_{yf} \sin \delta$$

$$m(\dot{v}_y + v_x r) = F_{yf} \cos \delta + F_{yr}$$

$$I_z \dot{r} = I_f F_{yf} \cos \delta - I_r F_{yr}, \quad \dot{\psi} = r$$

$$\alpha_f = \delta - \arctan \frac{v_y + I_f r}{v_x}, \quad \alpha_r = -\arctan \frac{v_y - I_r r}{v_x}$$